Warung Pintar: Aiming to digitize millions of family-owned stalls and small shops with Google Cloud

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About Warung Pintar

Warung Pintar is a technology company that provides solutions to help warungs—stalls or small shops—to order products and monitor their business progress. The business started operations in 2017 and had raised over $35 million from investors by early 2019. It aims to become the gold standard for micro-businesses in Indonesia.

Industries: Technology

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Location: Indonesia

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BigQuery (https://cloud.google.com/bigquery/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Cloud SQL (https://cloud.google.com/sql/)

Cloud Composer (https://cloud.google.com/composer/)

Cloud Storage (https://cloud.google.com/storage/)

Looker Studio (https://marketingplatform.google.com/about/data-studio/)

Container Registry (https://cloud.google.com/container-registry/)

Virtual Private Cloud (https://cloud.google.com/vpc/)

Cloud IoT Core (https://cloud.google.com/iot-core/)

AI Platform (https://cloud.google.com/ai-platform/)

TensorFlow (https://www.tensorflow.org/)

Istio (https://cloud.google.com/istio/)

Google Cloud results

- Builds a data-driven business optimized to help warungs grow and succeed
- Improves overall efficiency in development cycles

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- Benefits from responsive, technically skilled support
- Gains opportunity to evaluate advanced IoT, deep learning, AI, and machine learning services

Family-owned stalls and small shops known as warungs are vibrant contributors to Indonesia's economy and society. From gado gado (a peanut sauce-dressed salad of vegetables, boiled eggs, tofu, potatoes, and rice) to beef rendang, from ice-cold drinks to toothpaste, warungs typically sell food or essential day-to-day items to locals and visitors. However, until recently, financial and resource constraints made accessing new technologies or services difficult for warung owners and operators. Thanks to Warung Pintar, this is changing.

Warung Pintar (https://warungpintar.co.id/) started in 2017 after early-stage venture capital firm East Ventures—backer of prominent Indonesian businesses such as Tokopedia, Traveloka, and Mercari—opened a co-working space in Jakarta Smart City, JSCHive.

East Ventures' management team met with the owner of a warung that operated on the pedestrian sidewalk in front of Jakarta Smart City and decided to help his business. Jakarta Smart City relocated the warung into the JSCHive complex parking area and used technologies from its portfolio to renovate the micro-business.

One month after the renovations, the warung owner advised East Ventures that he had increased his income more than seven times.

Market to millions of warungs

The East Ventures team realized it could develop a digitization product to be marketed to millions of warungs across the country. The team continued to size up the market from the bottom up and consider how to scale the product, using venture capital models. Agung Bezharie Hadinegoro, then an Associate at East Ventures, established an operations team, while colleague Willson Cuaca focused on fundraising.

Cuaca now chairs Warung Pintar, while Hadinegoro is Chief Executive Officer. Working with Chief Operating Officer and Co-Founder Harya Putra, Chief Technology Officer and Co-Founder Sofian Hadiwijaya, and VP of Business and Co-Founder, Christian Winata, Cuaca and Hadinegoro identified the key issues facing warung owners as:

- **An inability to view and manage expenses, profits, and inventory**

- **Extended distribution chains that add cost and inefficiency**
- **Uncertainty over product availability and supply**

Warung Pintar created products to address these issues, including a system to record and manage point-of-sale transactions; a system to monitor, report on and analyze supply chain performance; and an app to enable warung owners and operators to order and track items, as well as monitor performance from a single dashboard.

A cultural touchstone

From its foundation, Warung Pintar knew its business was engaging a sector that was a cultural touchstone and part of Indonesians' daily lives. It identified social contribution as a key value and shouldered the responsibility of transforming warungs to help support Indonesia's economic development. "We aim to be the gold standard for servicing this critical sector," says Agung Julisman, VP of Engineering at Warung Pintar.

Warung Pintar also realized developing and executing the right technology strategy would be integral to its success. The business considered operating infrastructure in an on-premises data center, but realized it would have to assign team members to hardware maintenance and management rather than innovation and product development. It then decided to start operating in the cloud.

Infrastructure and smart analytics

After initially deploying on a traditional cloud, Warung Pintar decided it needed a cloud service with the infrastructure and smart analytics services that could help deliver:

- Warehouse inventory maintenance and management, including providing its procurement team with real-time insights on stock purchase requirements and consumption trends
- Insights into supply-chain management and distribution
- Use of the Internet of Things in combination with a point-of-sale system and CCTV to notify delivery fleets about warung opening times
- Identification of the easiest routes for deliveries
- Insight into customers to be able to make personalized recommendations
- Identification of the number of people at a warung at any given time to identify busy periods as well as demand patterns and customer demographics

Warung Pintar decided Google Cloud (https://cloud.google.com/) was best positioned to support its vision. The business began using the platform in mid-2019 and completed most of the installation—without engaging a partner—by the end of the quarter. “We received full assistance from the Google Cloud team, specifically for the system’s implementation, and the overall process was very smooth,” says Julisman. The business plans to go all in on Google Cloud once it works around vendor lock-in on its existing cloud.

“We use a range of Google Cloud services—including Google Kubernetes Engine to orchestrate the containers that automate our application scaling and management—and these provide the ease of use, simplicity, and management that enable our team to focus on our business goals."

—Sofian Hadiwijaya, Co-Founder and Chief Technology Officer, Warung Pintar

Ease of use, simplicity, and management

Warung Pintar now supports five business units—Core Product, Business Acceleration, Distribution, Merchant, and Hack Tribe—with a centralized platform from Google Cloud. "We use a range of Google Cloud services—including Google Kubernetes Engine (/kubernetes-engine) to orchestrate the containers that automate our application scaling and management—and these provide the ease of use, simplicity, and management that enable our team to focus on our business goals," says Hadiwijaya.

These other services include Istio (/istio) to connect, monitor, and manage its microservices architecture; Cloud SQL (/sql) to manage and run its relational databases; Cloud Composer (/composer) to author, schedule, and monitor pipelines across clouds; and a BigQuery (/bigquery) data warehouse that combines with Cloud Storage (/storage) to support data processing and analysis, and Looker Studio (https://marketingplatform.google.com/about/data-studio/) to help visualize analysis and reports as graphs and tables.

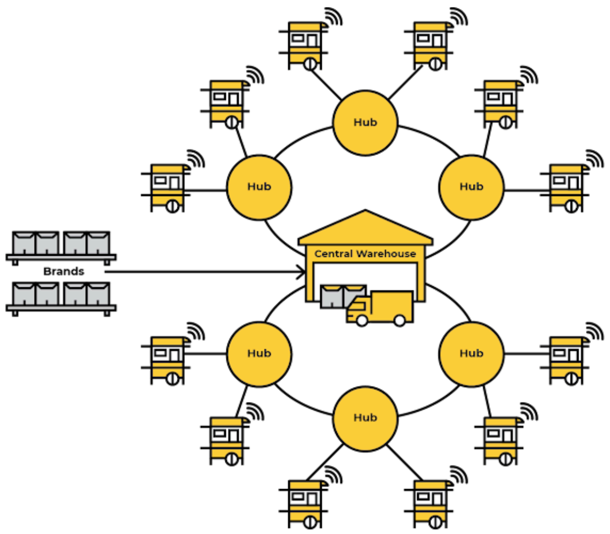
In addition, Container Registry (/container-registry) manages and controls access to Docker images and supports direct deployment to Google Kubernetes Engine. A custom Virtual Private Cloud (/vpc) (VPC) provides connectivity for Google Kubernetes Engine clusters and other resources.

A greater role than anticipated

With Google Cloud, Warung Pintar is meeting all its availability requirements and significantly reducing latency, as well as improving the visibility of its cloud infrastructure, data, and applications. “Google Cloud is playing a greater role in our sustainability as a business than we originally anticipated,” says Julisman. “It allows us to improve transparency, provide greater assurance around product orders and deliveries, and increase the effectiveness and efficiency of product development and growth planning. We are driving these improvements in product development and planning by maximizing analytics and making data-led decisions.

“We are also using blue-green deployments (operating two identical production environments, one of which is live) and canary deployments (initially rolling out releases to subsets of infrastructure before deploying more widely) to ultimately build a data-driven business that increases the efficiency of traditional warungs in Indonesia."

Warung Pintar's system now accommodates 30 to 40 orders of up to 500 items each per "hub," larger warungs that operate as extensions of its central warehouse and enable last-mile partners to deliver to warungs. "We found this very advantageous for our partners," says Julisman.



Warung Pintar hub delivery model

The Google Cloud team continues to support Warung Pintar, and the business is pleased with account and technical personnel's deep cooperation, prompt feedback, and high-quality deliverables. "We are excited by the opportunities to forge an even closer relationship with Google Cloud when it opens a new region in Jakarta this year," says Julisman. "The decision by Google Cloud encourages us to explore and use other Google Cloud services in future, as our technology strategy is based heavily on the platform."

> "Overall, we are extremely excited by the potential of Google Cloud to continue building our business domestically in a market of over 3 million warungs."
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> *—Agung Julisman, VP of Engineering, Warung Pintar*

IoT, AI Platform, and TensorFlow under consideration

Warung Pintar is looking into the potential of services such as IoT Core (/iot-core) that can help it connect, manage, and ingest data from millions of devices, AI Platform (/ai-platform) to build AI applications running on Google Cloud; and TensorFlow (https://www.tensorflow.org/) to create machine learning and deep learning models.

“Overall, we are extremely excited by the potential of Google Cloud to continue building our business domestically in a market of over 3 million warungs,” says Julisman.

Learn More

Reach out to our team to see how Google Cloud can help your business.

(https://cloud.google.com/